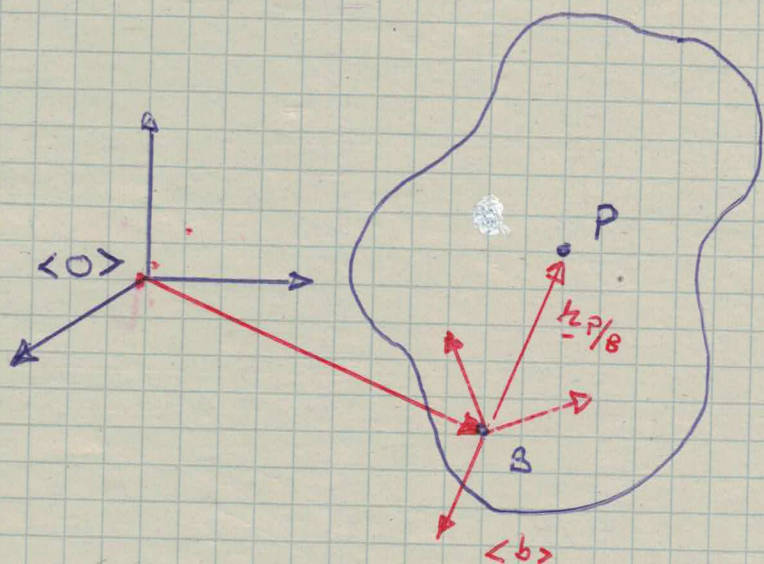


RIGID BODY DYNAMICS

1/

RIGID BODY A body is rigid if the distance between any two points is constant



$$\forall P, B \quad |P-B| = |\underline{r}_{P/B}| = \text{const.} \quad \forall t$$

$$\underline{v}_{P/O} = \underline{v}_{B/O} + \underline{\omega}_{B/O} \times \underline{r}_{P/B}$$

$$\underline{a}_{P/O} = \underline{a}_{B/O} + \dot{\underline{\omega}}_{B/O} \times \underline{r}_{P/B} + \underline{\omega}_{B/O} \times (\underline{\omega}_{B/O} \times \underline{r}_{P/B})$$

CENTER OF MASS

$$\underline{r}_{C/O} \triangleq (C-O) \triangleq \frac{\int_V (\underline{r}-O) \rho(\underline{r}) dV}{\int_V \rho(\underline{r}) dV}$$

total mass
of the body

$\rho(\underline{r})$ density
function

Properties

2/

$$\begin{aligned}
 \bullet \quad \underline{z}_{c/o} &\stackrel{\Delta}{=} (C-O) = \frac{\int_V [(P-B) + (B-O)] g(P) dV}{m} = \\
 &= \frac{\int_V (P-B) g(P) dV}{m} + \frac{\int_V (B-O) g(P) dV}{m} = \\
 &= (C-B) + (B-O) \frac{\int_V \cancel{g(P)} dV}{\cancel{m}} \overset{m_2}{=} \underline{z}_{c/s} + \underline{z}_{s/o}
 \end{aligned}$$

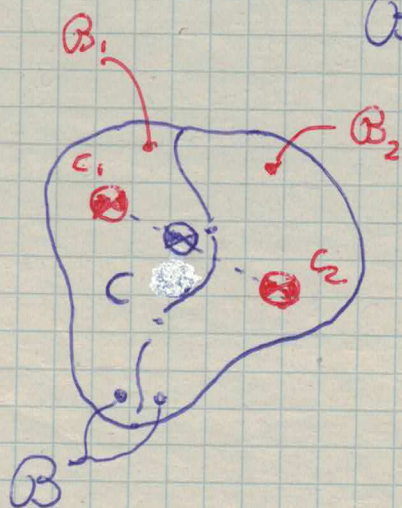
- Considers a body B and assume it can be "split" into 2 (or more parts) B_1, B_2 s.t.

$$B = B_1 \cup B_2$$

(the whole is the mm of the parts)

$$B_1 \cap B_2 = \emptyset$$

(the parts are distinguished)



$$(C-O) = \frac{\int_V (P-O) g(P) dP}{m} =$$

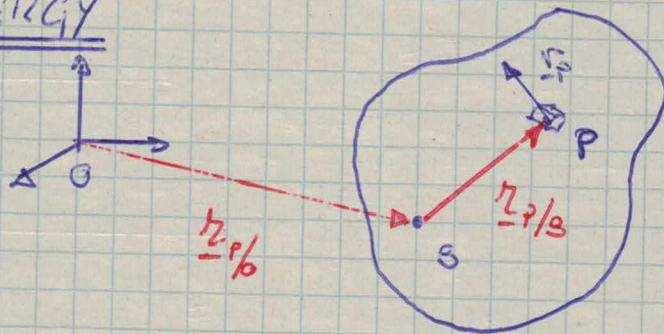
$$= \frac{\int_{V_1} (P-O) g(P) dV_1 + \int_{V_2} (P-O) g(P) dV_2}{m}$$

$$= \frac{m_1 (C_1-O) + m_2 (C_2-O)}{m_1 + m_2}$$

$$\bullet \quad (G-C_1) \times (G-C_2) = 0 \quad (*)$$

(*) hint compute $(C-C_1)$ and $(C-C_2)$

KINETIC ENERGY



$$m = \int_V \underbrace{\rho(P) dV}_{dm}$$

$$dT = \frac{1}{2} \underline{v}_{P/B} \cdot \underline{v}_{P/B} dm$$

$$T = \frac{1}{2} \int_V \underline{v}_{P/B} \cdot \underline{v}_{P/B} \rho(P) dV = \frac{1}{2} \int_V [\underline{v}_S + \underline{\omega}_B \times \underline{r}_{P/B}] \cdot [\underline{v}_S + \underline{\omega}_B \times \underline{r}_{P/B}] \rho dV$$

$$\underline{v}_{P/B} = \underline{v}_{S/B} + \underline{\omega}_B \times \underline{r}_{P/B}$$

Notice that:

$$\begin{aligned} \underline{v}_{P/B} \cdot \underline{v}_{P/B} &= [\underline{v}_{S/B} + \underline{\omega}_B \times \underline{r}_{P/B}] \cdot [\underline{v}_{S/B} + \underline{\omega}_B \times \underline{r}_{P/B}] = \\ &= \underline{v}_{S/B} \cdot \underline{v}_{S/B} + 2 \underline{v}_{S/B} \cdot (\underline{\omega}_B \times \underline{r}_{P/B}) + (\underline{\omega}_B \times \underline{r}_{P/B}) \cdot (\underline{\omega}_B \times \underline{r}_{P/B}) \end{aligned}$$

Then

$$\begin{aligned} T &= \frac{1}{2} \int_V \underline{v}_{S/B} \cdot \underline{v}_{S/B} \rho(P) dV + \frac{1}{2} \int_V \cancel{2} \underline{v}_{S/B} \cdot (\underline{\omega}_B \times \underline{r}_{P/B}) \rho(P) dV + \\ &\quad + \frac{1}{2} \int_V (\underline{\omega}_B \times \underline{r}_{P/B}) \cdot (\underline{\omega}_B \times \underline{r}_{P/B}) \rho(P) dV = \\ &\quad \underline{\omega}_B \cdot \left[\underline{r}_{P/B} \times (\underline{\omega}_B \times \underline{r}_{P/B}) \right] \end{aligned}$$

triple product
permutation
property

$$= \frac{1}{2} m \underline{v}_{B/G} \cdot \underline{v}_{B/G} + \underline{v}_{B/G} \cdot \left[\underline{\omega}_{B/G} \times \underbrace{\int_V \underline{r}_{P/B} \rho(P) dV}_{m \underline{r}_{C/B}} \right] +$$

$$+ \frac{1}{2} \underline{\omega}_{B/G} \cdot \underbrace{\int_V \underline{r}_{P/B} \times (\underline{\omega}_{B/G} \times \underline{r}_{P/B}) \rho(P) dV}_{\underline{I}_B(\underline{\omega}_{B/G})}$$

INERTIA OPERATOR "inertia operator"

Properties of $\underline{I}_B(\underline{m})$

- $\underline{I}_B(\underline{m})$ is symmetric (and hence linear)

Proof $\underline{v} \cdot \underline{I}_B(\underline{m}) = \underline{v} \cdot \int_V \underline{r} \times (\underline{m} \times \underline{r}) \rho(P) dV =$

$$= \int_V \underbrace{\underline{v} \cdot \underline{r} \times (\underline{m} \times \underline{r})}_{(\underline{m} \times \underline{r}) \cdot (\underline{v} \times \underline{r})} \rho(P) dV = \underline{m} \cdot \underline{I}_B(\underline{v})$$

$$= \int_V \underbrace{[\underline{r} \times (\underline{v} \times \underline{r})]}_{[\underline{r} \times (\underline{v} \times \underline{r})]} \cdot \underline{m} \rho(P) dV$$

- $\underline{I}_C(\underline{m}) \triangleq \int_V \underline{r}_{P/C} \times (\underline{m} \times \underline{r}_{P/C}) \rho(P) dV$



centroidal
inertia operator

• $I_B(\underline{m}) = I_C(\underline{m}) + m \underline{r}_{c/B} \times [\underline{m} \times \underline{r}_{c/B}]$ transposition formula

Proof

$$\underline{I}_B(\underline{m}) = \int_V \underline{r}_{P/B} \times \left(\underline{r}_{P/C} \times \underline{r}_{P/B} \right) \rho(P) dV =$$

$\uparrow$ $\underline{r}_{P/C} + \underline{r}_{c/B}$ constat $\uparrow$ $\underline{r}_{P/C} + \underline{r}_{c/B}$ constat

$$= \int_V \underline{r}_{P/C} \times (\underline{m} \times \underline{r}_{P/C}) \rho dV + \int_V \underline{r}_{P/C} \times (\underline{m} \times \underline{r}_{c/B}) \rho dV +$$

$$+ \int_V \underline{r}_{c/B} \times (\underline{m} \times \underline{r}_{P/C}) \rho dV + \underbrace{\int_V \underline{r}_{c/B} \times (\underline{m} \times \underline{r}_{c/B}) \rho dV}_{m \cdot \underline{r}_{c/B} \times (\underline{m} \times \underline{r}_{c/B})}$$

$$= I_C(\underline{m}) + \left(\int_V \underline{r}_{P/C} \rho dV \right) \times (\underline{m} \times \underline{r}_{c/B}) +$$

$\underline{r}_{c/C} = \underline{0}$

$$+ \underline{r}_{c/B} \times \left(\underline{m} \times \int_V \underline{r}_{P/C} \rho dV \right) + m \underline{r}_{c/B} \times (\underline{m} \times \underline{r}_{c/B})$$

- 5.1/
- considers a body B and assume it can be "split" into 2 (or more) parts B_1, B_2 s.t.

$$B = B_1 \cup B_2$$

$$B_1 \cap B_2 = \emptyset$$

then

$$I_B(\underline{u}) = I_B^{(1)}(\underline{u}) + I_B^{(2)}(\underline{u})$$

← obvious since the integral is additive.

↑
inertia operators are additive

- $\underline{u} \cdot I_B(\underline{u}) > 0 \quad \forall \underline{u} \neq \underline{0}$